

Tarea 2 (Ejercicio 1)

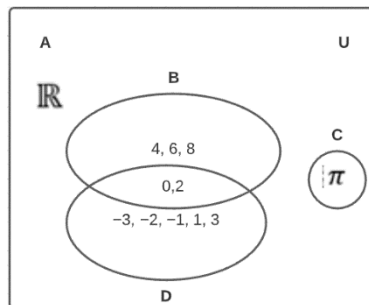
♥ Conjuntos ♥

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Ejercicio 1. (100 puntos)

Sean $A = \mathbb{R}$, $B = \{0, 2, 4, 6, 8\}$, $C = \{\pi\}$, $D = \{-3, -2, -1, 0, 1, 2, 3\}$ y $U = \mathbb{R}$.



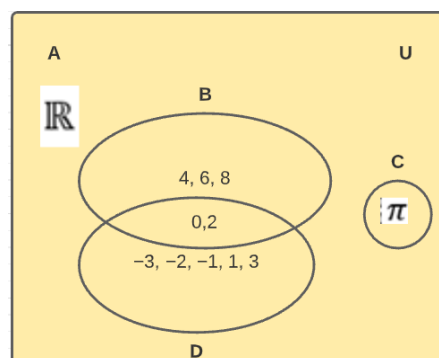
Encuentra:

1. $A \cup (C \cap D)$
2. $(A \cup B) - (C \cap D)$

1. $A \cup (C \cap D).$

$$A \cup (C \cap D) = \mathbb{R} \cup (\{\pi\} \cap \{-3, -2, -1, 0, 1, 2, 3\}) = \mathbb{R} \cup \emptyset = \mathbb{R}$$

$$\therefore A \cup (C \cap D) = \mathbb{R}$$



$$2. (A \cup B) - (C \cap D)$$

$$(A \cup B) - (C \cap D) = (\mathbb{R} \cup \{0, 2, 4, 6, 8\}) - (\{\pi\} \cap \{-3, -2, -1, 0, 1, 2, 3\})$$

$$\Rightarrow \mathbb{R} - \emptyset = \mathbb{R}$$

$$\therefore (A \cup B) - (C \cap D) = \mathbb{R}$$

